

IBRAHIM ALI GHANIM

MECHATRONICS ENGINEER | CONTROL SYSTEMS, SCADA & INDUSTRIAL AUTOMATION

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate (2026), ranked 9th out of 90 students (Top 10%), focused on PLC control, SCADA/HMI, electrical control, and industrial automation. Project experience with Siemens S7-1200/S7-200, Schneider Zelio, TIA Portal, WinCC, Ignition 8.3, Node-RED, Factory I/O, and Modbus TCP, covering sequential control, motor and pump control, interlocks, alarms, industrial communication, and AI-assisted fault detection.

EDUCATION

B.Sc. in Mechatronics Engineering | University of Baghdad - Al-Khwarizmi College of Engineering | June 2026
Ranked 9th out of 90 Mechatronics Engineering students (Top 10% of the cohort)

TECHNICAL SKILLS

PLC & Industrial Control: Siemens S7-1200/S7-200, Schneider Electric Zelio, TIA Portal, PLCSIM, Ladder Logic, Timers/Counters, Set/Reset, Comparators, Analog I/O & Scaling, Interlocks, Sequential Control, Motor/Pump/Tank Control, PID Fundamentals

SCADA, HMI & IIoT: Ignition 8.3, Siemens WinCC, Node-RED, Factory I/O, OPC UA, Tags, Alarming, Tag Historian, Trends, Event Logging, Manual/Auto Control, Fault Reset, E-STOP, Vision & Perspective Fundamentals

Industrial Communication: Modbus TCP, OPC UA, Ethernet/IP, MQTT, REST/JSON, Serial Communication, PLC-HMI Integration, Industrial Data Exchange

Electrical Control & Instrumentation: Power & Control Schematic Reading, Relays, Contactors, Circuit Breakers, Motor Starters, Industrial Sensors, Control Panels, Calibration & Troubleshooting, Basic AutoCAD Electrical Drawings

Control Engineering & Fluid Power: MATLAB/Simulink, Digital Control, Stability Analysis, PID Fundamentals, Pneumatic & Hydraulic Systems, Directional/Pressure/Flow Control Valves

AI for Automation: Python, FastAPI, Isolation Forest, AI-Assisted Anomaly & Fault Detection

CERTIFICATIONS

[Ignition 8.3 Credential](#) | Inductive University | 2026

[One Million Prompts](#) | Dubai Future Foundation | 2026

INTERSHIPS & TECHNICAL EXPERIENCE

SCADA & Electrical Systems Intern | Ministry of Electricity - Rusafa Directorate, Baghdad | Jul 2025 - Aug 2025

Observed SCADA control-room monitoring, alarm response, server infrastructure, electrical control panels, and field-to-control-room integration.

Instrumentation & Maintenance Trainee | Alchemy Scientific, Baghdad | Jun 2026 - Jul 2026

Performed calibration, functional testing, basic troubleshooting, probe/electrode care, and maintenance of laboratory and water-quality instruments.

SELECTED PROJECTS

Integrated PLC-HMI-IIoT Platform | System Integration | [View Project](#)

Integrated Siemens PLCSIM, WinCC HMI, Node-RED, Factory I/O, and VS Code in one synchronized environment for real-time control, monitoring, and industrial data exchange without external PLC hardware.

PetroSafe AI - Tank Safety & Fault Detection | Industrial Automation Prototype | [View Project](#)

Developed tank control and safety logic using Node-RED, Factory I/O, Modbus TCP, Auto/Manual control, E-STOP, Process Trip, fault latching, event logging, and FastAPI-based anomaly detection.

Multi-Station Tank & Pump Control System | Process Automation Simulation | [View Project](#)

Designed a 16-tank, four-station automation simulation using Siemens PLC/HMI, Modbus TCP, Node-RED, Factory I/O, automatic pump logic, alarms, trends, and leak detection.

Automated Conveyor & Elevator System | Siemens S7-200 | [View Project](#)

Programmed Ladder Logic for sensor- and counter-based three-level material transfer, conveyor sequencing, floor sorting, and automatic elevator return.

LANGUAGES

Arabic - Native | English - Intermediate